

YadYar Lite

Phase 1 - Topic, Dataset, and Baseline

T-29 - At-Risk-Student Prediction Under Class Imbalance



Members: Mahdiah Tajbakhsh; Atena Molaee; Fatemeh Masoomi

Presentation order: Atena Molaee -> Fatemeh Masoomi -> Mahdiah Tajbakhsh

Abstract

This Phase 1 study narrows T-29 into an early-warning binary-classification problem: identify students who will later fail or withdraw using only information available during the first 30 course days. The project uses a self-curated subset of the Open University Learning Analytics Dataset (OULAD), limited to module AAA and presentation 2013J. The subset contains 383 students, including 105 At-Risk students (27.4%). A reproducible scikit-learn Logistic Regression Pipeline is implemented as the baseline. Phase 2 will emphasize At-Risk Recall and PR-AUC rather than Accuracy, because the positive class is less frequent and a missed at-risk student is more serious than an unnecessary alert.

Phase 1 objective

Define a small, defensible ML question; document the data; implement a simple baseline; and specify a realistic evaluation and error-analysis plan. Final held-out results are intentionally reserved for Phase 2.

1. Narrow Project Question

Research question

Can students who will later fail or withdraw from one online course presentation be identified using only information available during the first 30 course days?

The unit of analysis is one student registered in module AAA, presentation 2013J. The problem is binary supervised classification. The target is created from the final course outcome:

- At Risk (1): final_result is Fail or Withdrawn.
- Not At Risk (0): final_result is Pass or Distinction.

The baseline produces an estimated risk probability. A decision threshold can later convert that probability into an early-warning flag. The intended use is supportive human review, not automatic punishment, exclusion, or grading.

2. Why the Scope Is Feasible

- One course presentation keeps the dataset small enough to inspect and reproduce.
- The first-30-day window creates a meaningful early-warning task.
- The target is clear and already available in OULAD.
- Logistic Regression is fast, interpretable, and appropriate for an undergraduate baseline.
- The project focuses on model behaviour and error trade-offs rather than production deployment.

3. Expected Component Input and Output

Element	Example	Purpose
Input	Early activity, early scores, previous attempts, credits, student attributes	Represent information available by day 30
Output	risk_probability = 0.00-1.00	Rank students for possible support

4. Dataset Description

The Open University Learning Analytics Dataset (OULAD) contains linked tables describing course presentations, student characteristics, virtual-learning-environment activity, assessments, and outcomes. This project uses a self-curated AAA/2013J subset distributed with the submission.

File	Rows	Role in this project
studentInfo.csv	383	Student attributes and final outcome
studentVle.csv	180,982	VLE clicks by student, site, and course day
assessments.csv	6	Assessment metadata
studentAssessment.csv	1,633	Submission timing and scores

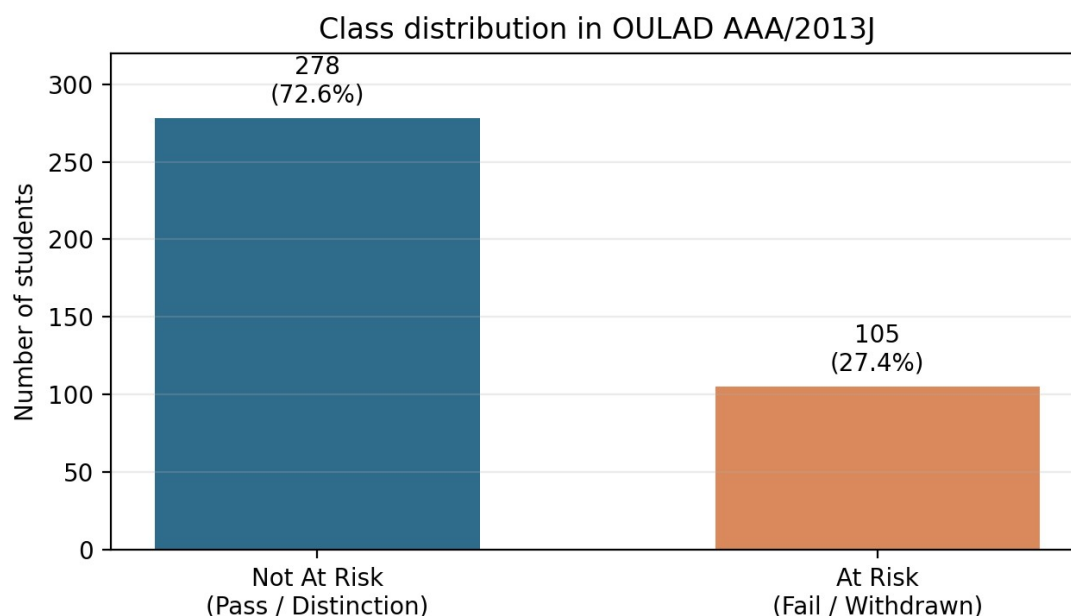


Figure 1. Class distribution in the selected OULAD subset.

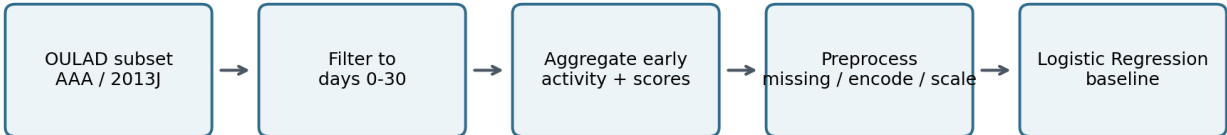
The minority class contains 105 students (27.4%). A classifier that always predicts Not At Risk could appear reasonably accurate while detecting no at-risk students. This motivates minority-focused evaluation in Phase 2.

5. Feature Plan and Leakage Prevention

Group	Features	Treatment
Static categorical	gender, region, highest_education, imd_band, age_band, disability	Most-frequent imputation + one-hot encoding
Static numeric	num_of_prev_attempts, studied_credits	Median imputation + standardization
Early VLE	total_clicks_30d, active_days_30d, unique_sites_30d, avg_clicks_per_active_day	Aggregated only from days 0-30
Early assessment	assessment_count_30d, avg_score_30d	Submitted only during days 0-30

Leakage controls

- final_result is used only to construct the target and is not passed to the model.
- VLE events are filtered to course days 0-30 before aggregation.
- Assessment submissions and scores are filtered to days 0-30.
- Imputation, encoding, and scaling are fitted inside the Pipeline using training data.
- No later clicks, later submissions, or final grades are used as predictors.



Final held-out evaluation, threshold analysis, and error inspection are deferred to Phase 2.

Figure 2. Reproducible Phase 1 data and baseline workflow.

6. Baseline Setup

The official baseline is scikit-learn Logistic Regression. It is deliberately simple: the model is fast to train, produces class probabilities, and is easier to explain in oral questioning than a heavily tuned nonlinear model.

Component	Configuration
Categorical preprocessing	SimpleImputer(most_frequent) -> OneHotEncoder(handle_unknown=ignore)
Numeric preprocessing	SimpleImputer(median) -> StandardScaler
Classifier	LogisticRegression(max_iter=1500, random_state=42)
Split	Stratified 80/20; 306 train and 77 reserved test students
Saved artifact	models/phase1_logistic_baseline.joblib

Reproducible execution

Windows

```
pip install -r requirements.txt
run_phase1.bat
```

Linux/macOS

```
pip install -r requirements.txt
bash run_phase1.sh
```

The script builds the engineered dataset, creates the fixed split, fits the preprocessing and Logistic Regression Pipeline, saves the model, and writes machine-readable setup summaries. It does not report the held-out Phase 2 evaluation in the Phase 1 report.

7. Phase 2 Evaluation Plan

Metric	Meaning	Why it fits T-29
At-Risk Recall	$TP / (TP + FN)$	Directly measures how many truly at-risk students are detected.
PR-AUC	Precision-Recall performance across thresholds	Focuses on ranking quality for the less frequent positive class.

Accuracy will be shown only as supporting context, because a majority-class model can obtain a seemingly acceptable Accuracy while producing zero At-Risk Recall.

Planned operating-point analysis

- Compare thresholds 0.50 and 0.30.
- Report TP, FP, FN, and TN using confusion matrices.
- Use the illustrative cost $5 \times FN + 1 \times FP$ to reflect asymmetric error importance.

Planned simple error analysis

- False Negatives: at-risk students who would not be flagged.
- False Positives: successful students who would receive an unnecessary alert.
- Engagement slices: low, medium, and high early-click groups.
- Representative cases: compare early clicks, active days, early scores, and previous attempts.

8. Limitations and Next Steps

The subset contains only 383 students from one presentation, so results may not generalize to other modules, institutions, or years. Demographic variables may encode historical inequalities and require fairness review before any real-world use. The day-30 snapshot cannot represent later personal or academic events. Phase 2 will perform the held-out evaluation, threshold comparison, error inspection, and lightweight demo. A simple alternative-model comparison may be added only as optional bonus work after the core baseline is complete.

References and acknowledgement

[1] YadYar Lite course project handout, Phase 1 requirements.

[2] Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2015). Open University Learning Analytics dataset. UCI Machine Learning Repository. DOI: 10.24432/C5KK69.

[3] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825-2830.

Generative AI assistance was used for code organization, debugging, report drafting, and documentation. The team remains responsible for understanding, checking, and defending all submitted work.